Project: Porewater SO4/Cl

Month Year Samples

2026-10-19

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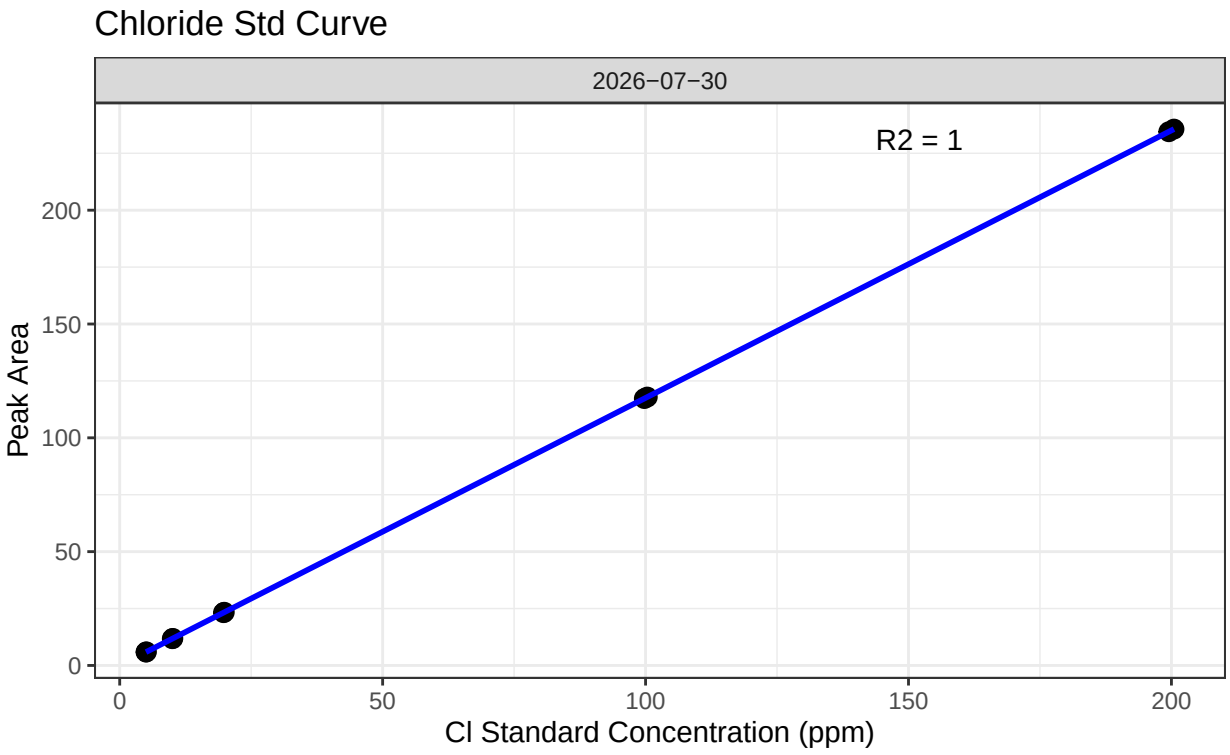
1 Run Information

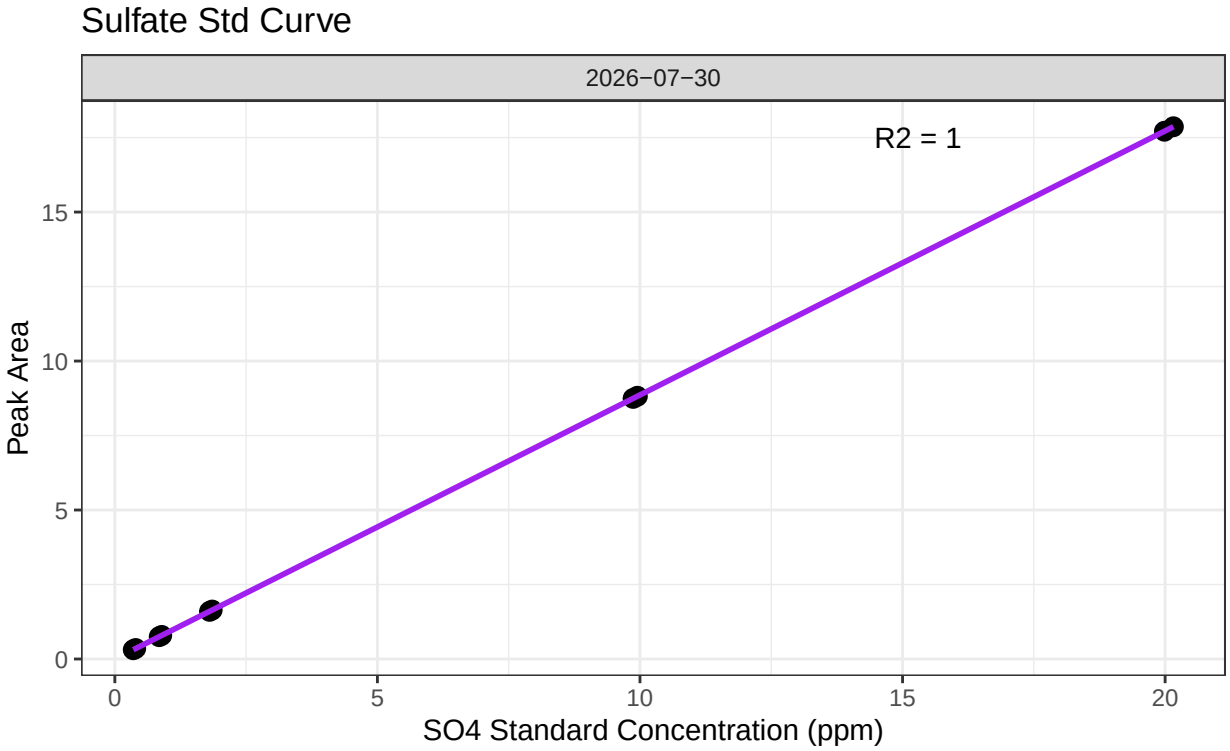
Run date: 2026-07-30
Run by: Zoe Read
Run notes: Run notes

SO4 file used: - CO2xN_202605_SO4.txt

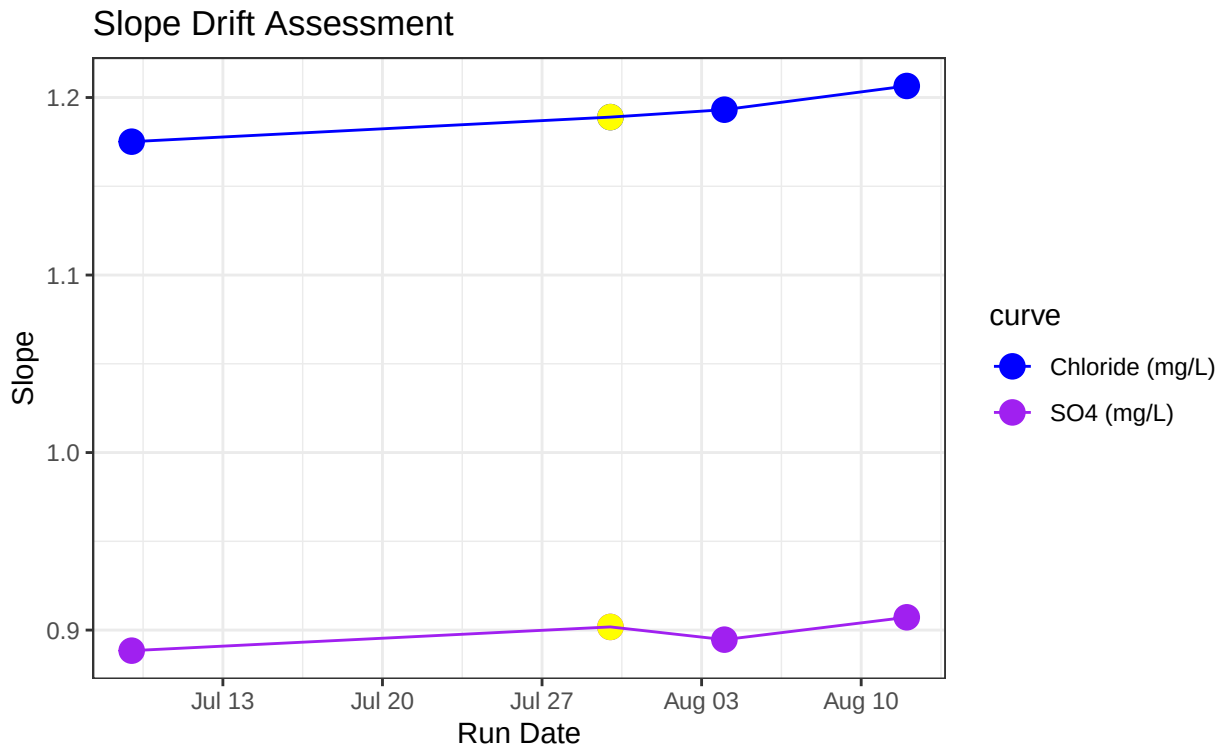
Cl file used: - CO2xN_202605_Cl.txt

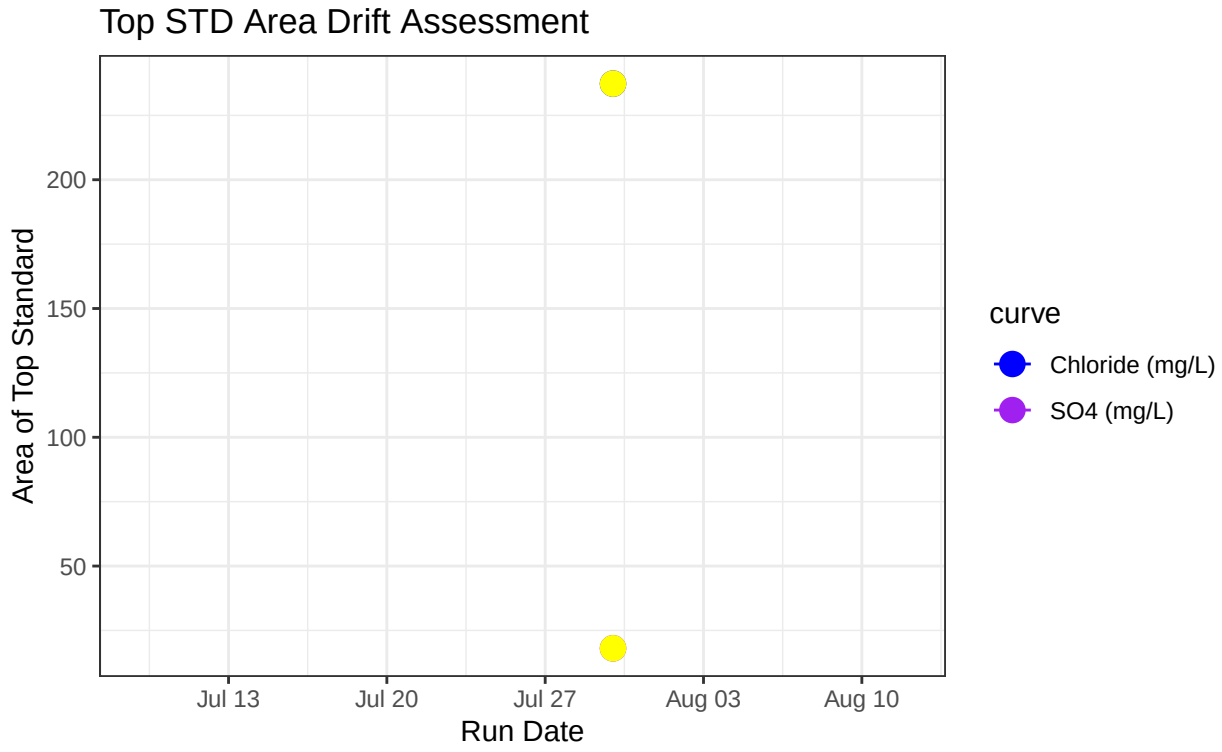
1.1 Assess Standard Curves






```
## [1] "QAQC log file exists and has been read into the code."
```






```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

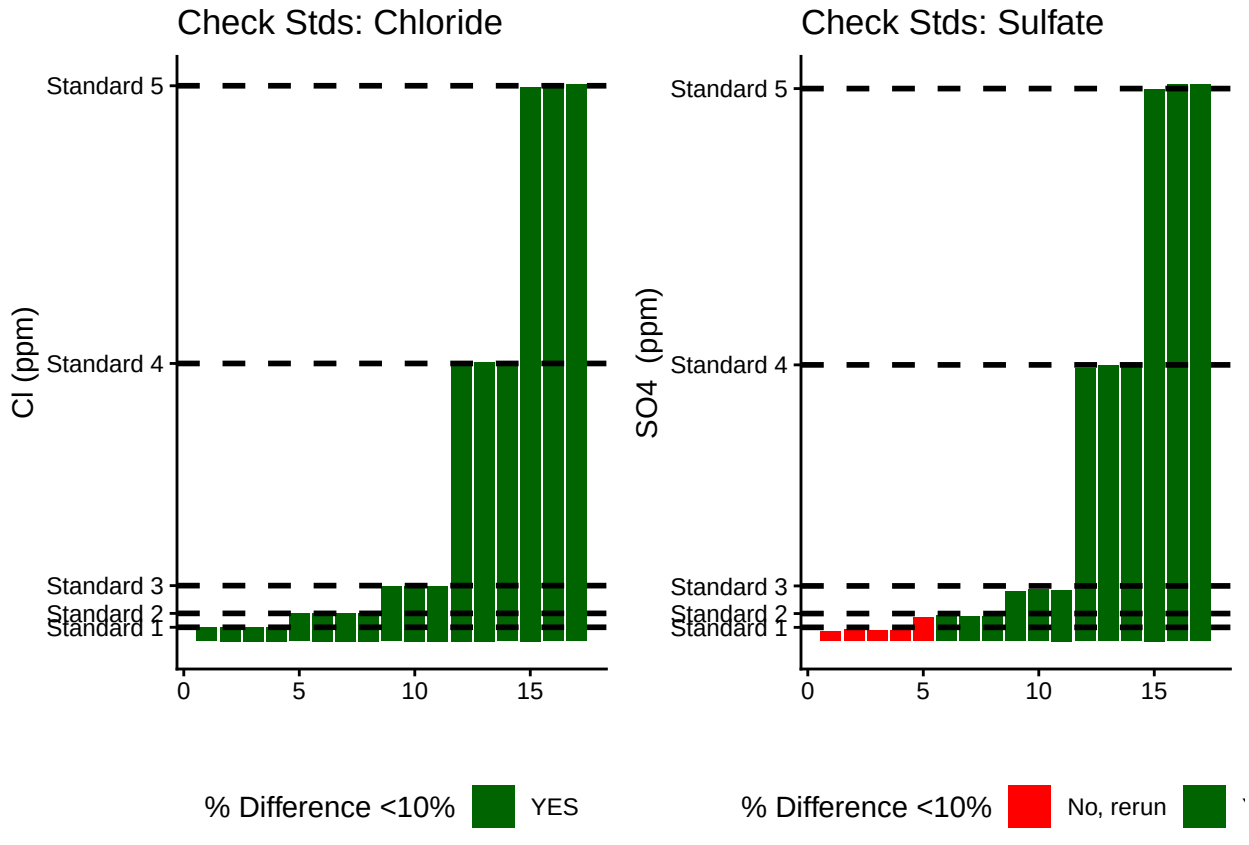
1.2 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID  mean_Cl  sd_Cl   cv_Cl flag_Cl
##   <chr>        <dbl>  <dbl>   <dbl> <chr>
## 1 Standard 1    5.06 0.0349 0.00691 Chloride Check Standard RSD within Range - ~
## 2 Standard 2   10.1  0.0512 0.00508 Chloride Check Standard RSD within Range - ~
## 3 Standard 3   19.8  0.0847 0.00427 Chloride Check Standard RSD within Range - ~
## 4 Standard 4  100.   0.372  0.00371 Chloride Check Standard RSD within Range - ~
## 5 Standard 5  200.   0.587  0.00293 Chloride Check Standard RSD within Range - ~
```

```
## # A tibble: 5 x 5
##   sample_ID  mean_SO4 sd_SO4  cv_SO4 flag_SO4
##   <chr>         <dbl>  <dbl>   <dbl> <chr>
## 1 Standard 1    0.392 0.0306 0.0780  Sulfate Check Standard RSD within Range - ~
## 2 Standard 2    0.891 0.0339 0.0380  Sulfate Check Standard RSD within Range - ~
## 3 Standard 3    1.84  0.0368 0.0200  Sulfate Check Standard RSD within Range - ~
## 4 Standard 4    9.93  0.0538 0.00542 Sulfate Check Standard RSD within Range - ~
## 5 Standard 5   20.1   0.0954 0.00475 Sulfate Check Standard RSD within Range - ~
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



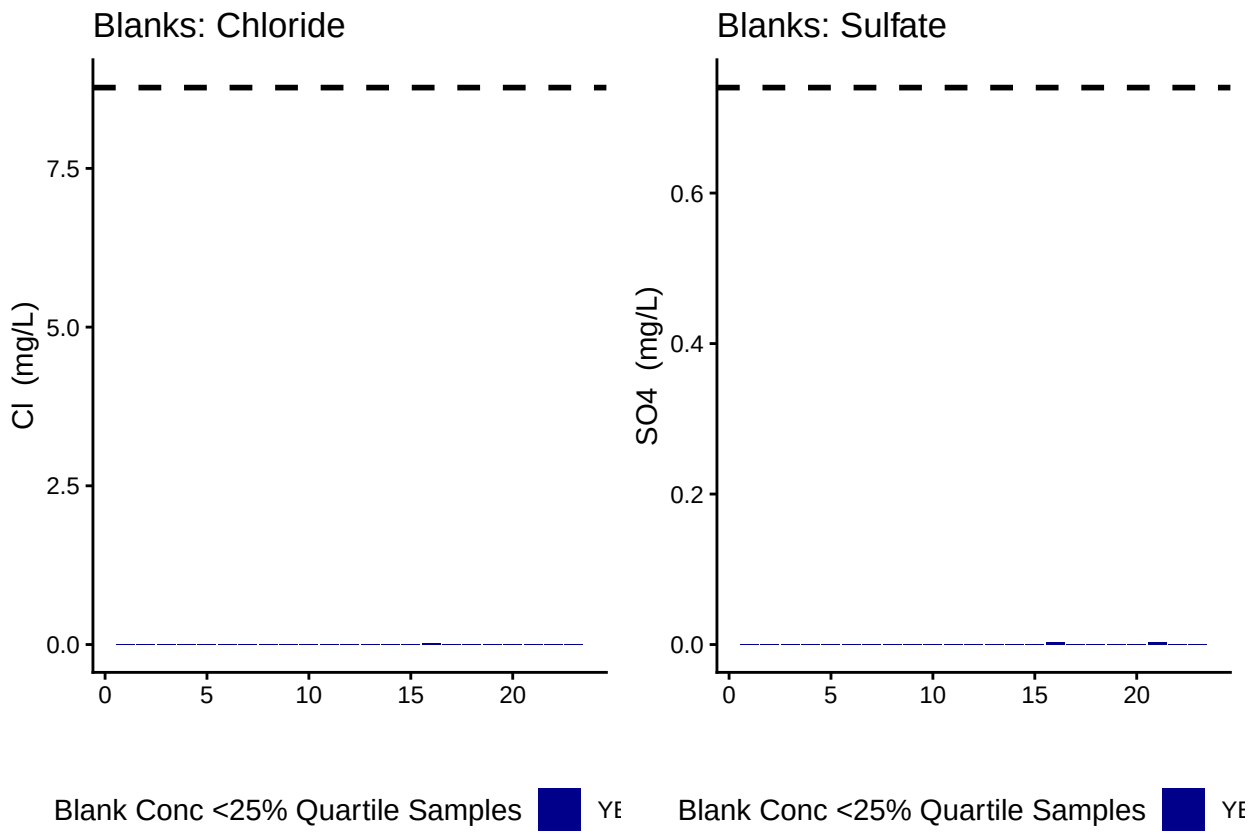

```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] "<80% of Sulfate Check Standards are within range of expected concentration - REASSESS"
```

1.3 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



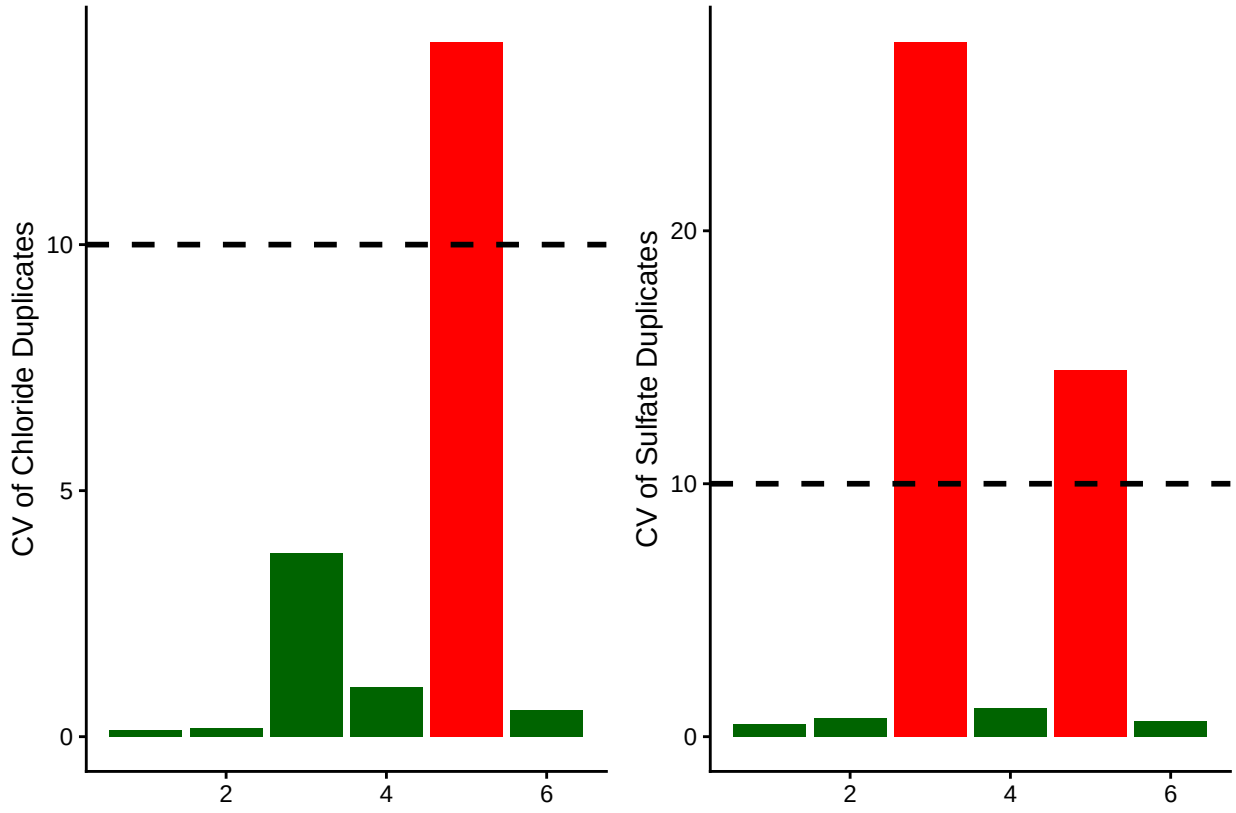

```
## Chloride blanks mean ppm:
```

```
## [1] 0.002308696
```

```
## Sulfate blanks mean ppm:
```

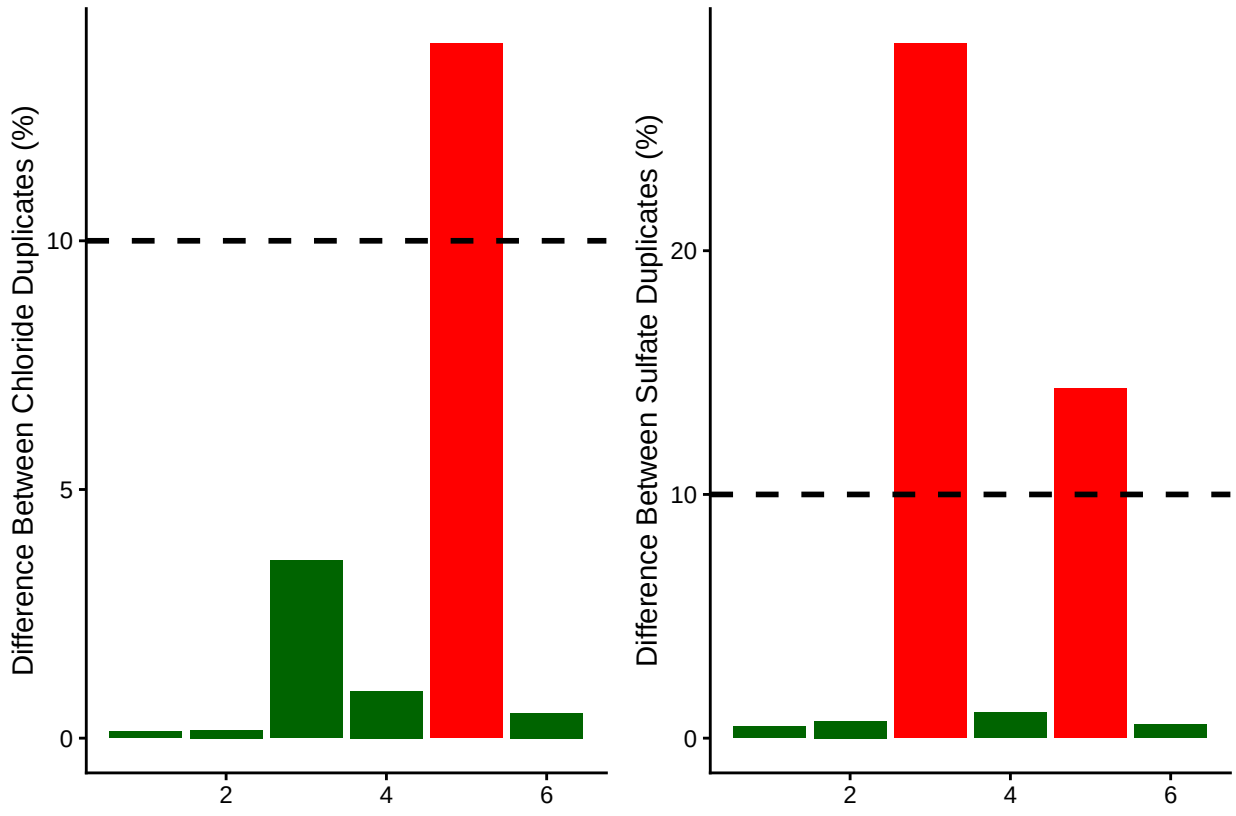
```
## [1] 0.0002913043
```

1.4 Assess Duplicates




```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] "<80% of Sulfate Duplicates have a CV <10% - REASSESS"
```

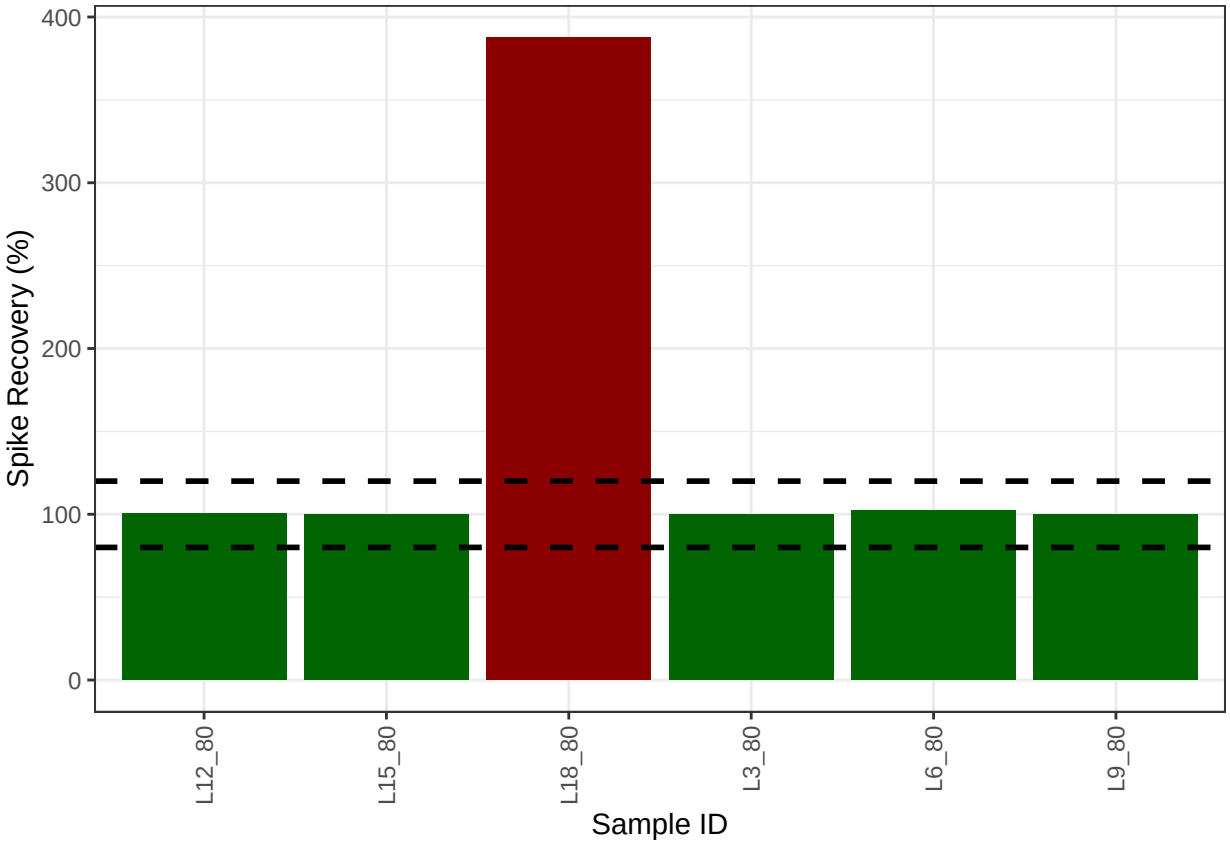



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] "<80% of Sulfate Duplicates have a CV <10% - REASSESS"
```

1.5 Calculate mmol/L concentrations & salinity, add dilutions

1.6 Assess Analytical Spikes




```
## [1] ">80% of SO4 spikes have a recovery between the high and low cutoff - PROCEED"
```

1.7 Check if samples are within the range of the standard curve

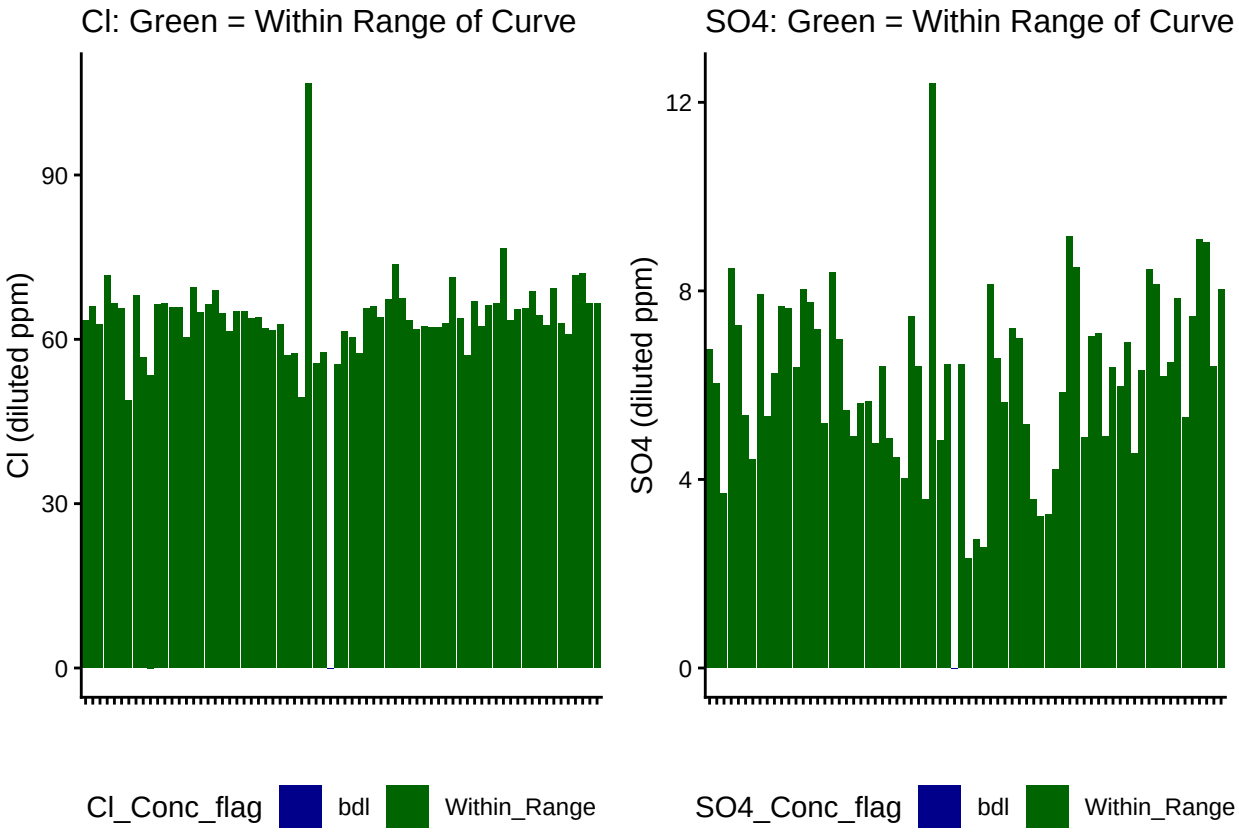


Table 1: SO4 samples

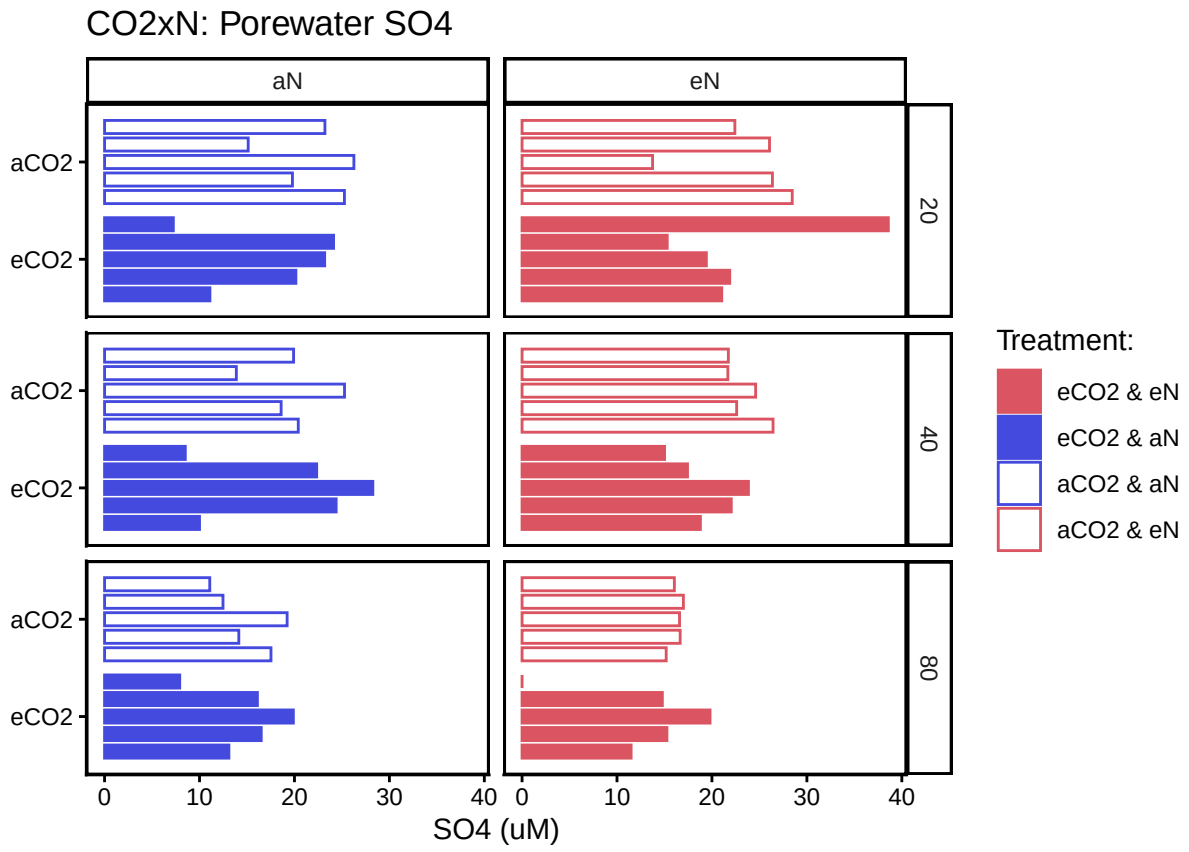
SO4_Conc_flag	Percent_samples
Within_Range	98.611111
bdl	1.388889

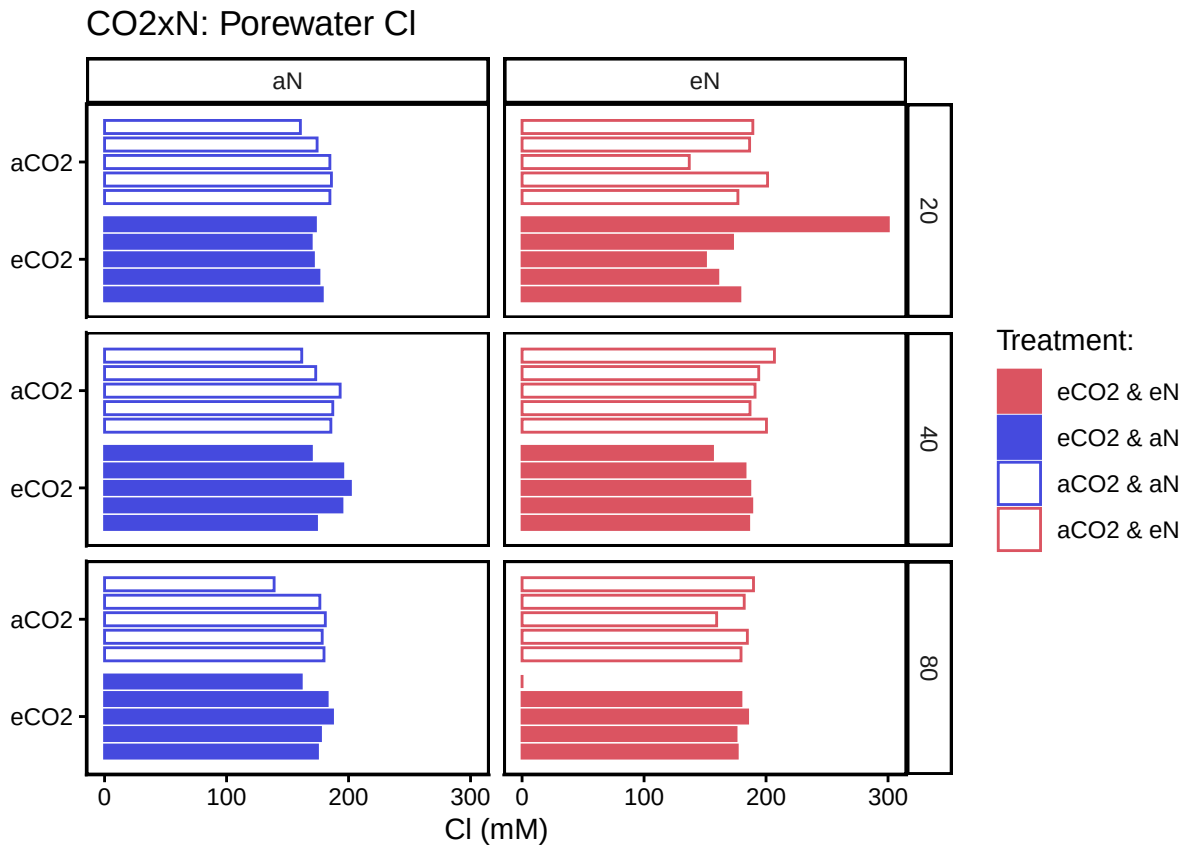
Table 2: Cl samples

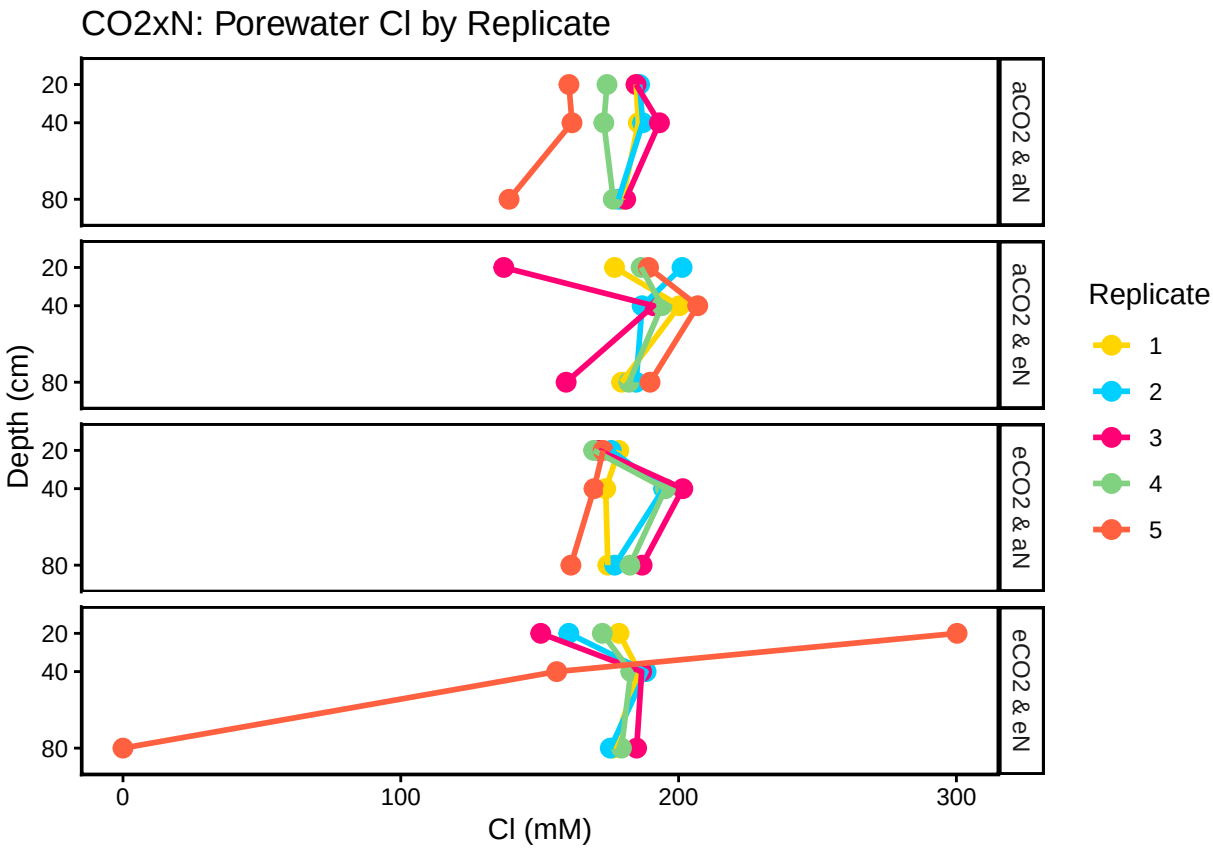
Cl_Conc_flag	Percent_samples
Within_Range	98.611111
bdl	1.388889

1.8 Check to see if samples run match metadata & merge info

```
## All sample IDs are present in metadata.
```


CO2xN: Porewater SO4
aN
eN
aCO2
eCO2
20
40
80
0
10
20
30
40
SO4 (uM)
Treatment:
eCO2 & eN
eCO2 & aN
aCO2 & aN
aCO2 & eN


CO2xN: Porewater Cl
aN
eN
aCO2
eCO2
20
40
80
0
100
200
300
Cl (mM)
Treatment:
eCO2 & eN
eCO2 & aN
aCO2 & aN
aCO2 & eN


CO2xN: Porewater Cl by Replicate
Depth (cm)
20
40
80
aCO2 & aN
aCO2 & eN
eCO2 & aN
eCO2 & eN
0
100
200
300
Cl (mM)
Replicate
1
2
3
4
5

#end